Chujun Zong

Dr.-Ing. · Life Cycle Assessment · Sustainable Building Design

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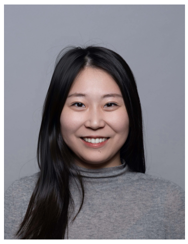

RESEARCH PROFILE

Dr.-Ing. (*magna cum laude*, TUM, 2026) specializing in dynamic life cycle assessment of the built environment, uncertainty modeling, and computational decision support. Developed GraphDLCA, a graph-database-driven framework for automated dynamic LCA, and MOSO, a multi-objective stochastic optimization framework for design under uncertainty. Author/co-author of 20+ peer-reviewed publications, including four first-author articles in JCR Q1 journals. Co-managed two federally funded research projects (BMWK, DFG), supervised 18 Master's theses, and received the Doce et Delecta Teaching Award (First Place, 2023).

EDUCATION

Dr.-Ing. Technical University of Munich **2022 – 2026**

Chair of Energy Efficient and Sustainable Design and Building. Supervisor: Prof. Werner Lang.
magna cum laude.

- Dissertation: Addressing Future Changes in Building Life Cycle Assessment: Uncertainty Modeling and Dynamic Life Cycle Assessment (DLCA) for Decision-Making.

M.Sc. Resource Efficient and Sustainable Building, Technical University of Munich **2019 – 2021**

- Thesis: Implementation of a Stochastic Methodology to Predict Window Opening Behavior in a Dynamic Building Simulation Program (published and presented at BauSim 2022).

B.A. Architecture, Technical University of Munich **2014 – 2019**

Exchange B.A. Architecture, École Polytechnique Fédérale de Lausanne (EPFL) **2016 – 2017**

ACADEMIC & PROFESSIONAL EXPERIENCE

Research Fellow & Doctoral Candidate — Technical University of Munich **2022 – 2026**

Chair of Energy Efficient and Sustainable Design and Building (Prof. Werner Lang).

- Co-managed two externally funded research projects (BMWK CircularGreenSimCity and DFG EarlyBIM2), with responsibilities for LCA methodology, web-tool development, workshops, deliverables, reporting, and coordination within interdisciplinary consortia.
- Developed and published two computational frameworks: GraphDLCA (Neo4j graph database for automated dynamic LCA) and MOSO (multi-objective stochastic optimization under uncertainty).
- Supervised 18 Master's theses (16 completed), four of which led to peer-reviewed conference publications with students as authors.
- Taught four courses in English and German at Bachelor and Master level (ca. 333 contact hours) and received Doce et Delecta Teaching Award, First Place, 2023.
- Mentored more than five student research assistants across teaching, research and administration, and contributed to the chair's recruitment of research associates.
- Maintained the chair's web presence, and coordinated institutional liaison with partner universities (South China University of Technology, TUM Asia).

Working Student — Siemens Energy AG **2020 – 2021**

Section of Construction, Design & Sustainability in Real Estate.

- Drafted the first corporate sustainability and workplace guidelines.
- Supported sustainable factory renovation projects with site and project managers, and developed workplace renovation plans for offices in Germany, Singapore, Malaysia, and Slovenia, including CAD drawings, material selection, and solution presentations.
- Presented work to the managing board and contributed to input sessions within the Real Estate department.

Independent consulting engagement on F-gas regulation and CBAM **2026 – now**

- Advisory work on EU F-gas Regulation and CBAM compliance for industrial clients.

FUNDED RESEARCH PROJECTS

CircularGreenSimCity **2022 – 2025**

Holistic life cycle assessment of urban districts. Funded by the German Federal Ministry for Economic Affairs and Climate Action (BMWK).

- Responsible for the LCA work package of archetype-based district stock modeling, scenario development and assessment methodology, and programmed core components of the district renovation web tool used by municipal planners in Würzburg and Asperg.
- Managed budget, milestone and final reporting, and organized consortium workshops across university, municipal and industry partners.
- Embedded five Master's theses in the project, feeding deliverables and three conference publications, and mentored four student research assistants.

EarlyBIM2 **2022 – 2024**

Knowledge database and machine learning assistance for performance-oriented building. Funded by the German Research Foundation (DFG).

- Developed early-design assessment methods under incomplete information, including benchmarking machine-learning surrogates against physics-based simulation.
- Responsible for milestone and final reporting, external communication with participating universities, regular onsite presentations, and collaborative publications (four joint publications in total).
- Initiated the open-source release of the EarlyData knowledge base, and mentored one student research assistant.

RESEARCH FRAMEWORKS & TOOLS

- GraphDLCA: graph-database structure and calculation framework for automated dynamic LCA of buildings (Neo4j/Cypher, Python, demonstration video on my personal homepage).
- MOSO / MOSO-II: multi-objective stochastic optimization framework coupling building energy simulation with embodied carbon objectives (NSGA-II, Monte Carlo, Python).
- Contributions to the CircularGreenSimCity district assessment web tool (Python).
- Open-source publication of EarlyData knowledge base from EarlyBIM2 (Zenodo: https://doi.org/10.5281/zenodo.10962087).

TEACHING EXPERIENCE

Student Supervision

18 Master's theses supervised (2023 – 2026, 16 completed, 2 ongoing). Four theses led directly to peer-reviewed international conference publications with students as authors, with two further journal manuscripts in preparation. Topics anchored in funded projects, doctoral research, and industry cooperation (e.g., Siemens Energy). One thesis grew into a student start-up.

System Effects and Interdependencies of Sustainable Planning in Civil Engineering **2025 – 2026**

Master's interdisciplinary seminar, English, co-lecturer, Technical University of Munich

Own seminar group. Ca. 17 contact hours, ca. 50 students per semester.

Introduction to Scientific Writing **2023 – 2026**

Bachelor's and Master's seminar, English, sole lecturer, Technical University of Munich

Ca. 97 contact hours, ca. 40 students per semester.

Sustainable Building & Ecological Building Basic Modules **2022 – 2026**

Bachelor's lecture and exercise, German, lead lecturer (exercises), teaching assistant (lectures), Technical University of Munich

Ca. 82 contact hours, ca. 200 students per semester. Mentored 1-2 student research assistants per semester.

Sustainable Lighting Technology **2022 – 2025**

Master's lecture, English, sole lecturer, Technical University of Munich

Ca. 137 contact hours, ca. 100 students per semester. Doce et Delecta Teaching Award (First Place), 2023. Mentored 1 student research assistant per semester.

HONORS & AWARDS

- Doce et Delecta Teaching Award, First Place — Department of Civil and Environmental Engineering, TUM (Winter Semester 2022/2023, Sustainable Lighting Technology)
- National Award for Outstanding Self-funded Students Abroad (Category A), 2024 – the national distinction for Chinese self-funded doctoral researchers abroad.

ACADEMIC SERVICE

- Peer reviewer for international journals including Building and Environment, Building Simulation, Sustainability, Scientific Reports, Discover Artificial Intelligence, Cell Reports Sustainability, among others.
- Conference presentations at BauSIM (2022), Building Simulation (2023, 2025), and WSBE (2024).
- Peer reviewer for international conferences including Building Simulation (2025) and WSBE (2024).
- Member of IBPSA-DACH (Germany, Austria, Switzerland), DGNB (German Sustainable Building Council) Registered Professional.

TECHNICAL SKILLS & LANGUAGES

- Programming and data (Python, Neo4j/Cypher, MATLAB), Building simulation (IDA ICE, EnergyPlus, TRNSYS), Architectural design and BIM toolchain.
- Languages: Chinese (native), German (professional), English (professional, IELTS 8.0), French (conversational).

SELECTED PUBLICATIONS (full publication list enclosed)

Peer-Reviewed Journal Articles

- **Zong, C.**, Banihashemi, F., Petzold, F. & Lang, W. (2026). Graph-based Database Structure and Calculation Framework for Automated Dynamic Life Cycle Assessment of Buildings. Accepted for publication in Automation in Construction. (JCR Q1)
- **Zong, C.**, Banihashemi, F., & Lang, W. (2025). Dynamic life cycle impact assessment (DLCIA) in a sustainable building planning process. Scientific Reports, 15(1), 32680. (JCR Q1)
- **Zong, C.**, Chen, X., Deghim, F., Staudt, J., Geyer, P., & Lang, W. (2024). A holistic two-stage decision-making methodology for passive and active building design strategies under uncertainty. Building and Environment, 251. (JCR Q1)
- **Zong, C.**, Margesin, M., Staudt, J., Deghim, F., & Lang, W. (2022). Decision-making under uncertainty in the early phase of building façade design based on multi-objective stochastic optimization. Building and Environment, 226, 109729. (JCR Q1)
- Napps, D., Staudt, J., Saluz, U., Chen, X., Steiner, D., **Zong, C.**, Deghim, F., Lang, W., Geyer, P., Schnellenbach-Held, M., Petzold, F., Borrmann, A., König, M. (2026). Evaluation of Building Design Variants in Early Phases on the Basis of Adaptive Detailing Strategies. Buildings, 16(4), 685. (JCR Q2)
- Banihashemi, F., Weber, M., Deghim, F., **Zong, C.**, & Lang, W. (2024). Occupancy modeling on non-intrusive indoor environmental data through machine learning. Building and Environment, 254. (JCR Q1)

Selected Conference Papers

- **Zong, C.**, & Lang, W. (2025). Dynamic factors in life cycle assessment: systematic review and categorization. Building Simulation Conference Proceedings 2025.
- **Zong, C.**, Lang, W., Pancheri, F. & Sun, Y. (2025). Assessing pedestrian level barriers using mobile robot to validate the 15-minute city concept. Building Simulation Conference Proceedings 2025.
- **Zong, C.**, Sun, Y., & Lang, W. (2024). System Dynamics Modeling of Life Cycle Carbon Footprints for Building Wall Insulation Materials. *IOP Conference Series: Earth and Environmental Science*, 1363.
- **Zong, C.**, Reitberger, R., Deghim, F., Staudt, J., Larkin, J. A., & Lang, W. (2023). A comparative study of dynamic building simulation and machine-learning within a two-stage multi-objective stochastic optimization framework. Building Simulation Conference Proceedings, 18, 1540–1547.
- **Zong, C.**, Banihashemi, F., Vollmer, M., & Lang, W. (2022). Implementation of occupant behaviour models for window control using co-simulation approach. BauSIM 2022.